Pregnancy Support System

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Project Proposal Report

Livera S.D – IT22638168

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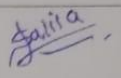
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Livera S.D	IT22638168	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr. Kapila Dissanayake		

ABSTRACT

Pregnancy is honestly one of the hardest times in any woman's life, not just physically but emotionally too. In Sri Lanka, we have a pretty good antenatal care system on the physical side, run through the Family Health Bureau and the network of government hospitals and MOH clinics. But the emotional and mental health side of pregnancy care is something that gets almost no attention at all in the local system. Pregnant women here go to their clinic once a month, get their blood pressure checked, get their haemoglobin tested, and go home. Nobody asks them how they are really feeling.

The mental health gap is real and the numbers back it up. Studies done right here in Sri Lanka, at hospitals like De Soysa and Sri Jayewardenepura, found that between 15 to 23 percent of pregnant women attending antenatal clinics had significant depression or anxiety symptoms. But almost none of them were identified because there is no screening for mental health in the standard Sri Lankan antenatal protocol. They just go home and manage it alone.

My part of this app is an AI chatbot designed specifically for pregnant women in Sri Lanka. It talks to them in both Sinhala and English, which matters a lot because a woman in Matara or Kurunegala is not going to feel comfortable chatting in English only. The chatbot does more than just answer questions though. While the woman is chatting, the app quietly takes a photo from the front camera every 30 to 40 seconds and uses a facial expression model to pick up on whether she looks stressed, sad, or okay. It also watches how she types, how long she takes to reply, and what kind of words she uses. All of these signals get combined to figure out her mood. Then the chatbot adjusts how it talks to her based on that. If she seems really upset, it becomes more gentle, it recommends some calming Sinhala music, and it gently suggests she talk to her doctor or midwife.

Everything is saved on her phone locally so she can go back and read old conversations or listen to saved music even without internet. That offline part is really important because not everyone in Sri Lanka has reliable connectivity, especially in rural areas.

Keywords: pregnancy mental health Sri Lanka, Sinhala AI chatbot, facial emotion recognition, antenatal depression Sri Lanka, maternal wellbeing app, offline pregnancy support

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	**Artificial Intelligence**
CNN	**Convolutional Neural Network**
NLP	**Natural Language Processing**
ML	**Machine Learning**
FER	**Facial Expression Recognition**
API	**Application Programming Interface**
UI / UX	**User Interface / User Experience**
SLIIT	**Sri Lanka Institute of Information Technology**
FHB	**Family Health Bureau, Sri Lanka**
MOH	**Medical Officer of Health**
TRCSL	**Telecommunications Regulatory Commission of Sri Lanka**
SDG	**Sustainable Development Goal**
LLM	**Large Language Model**
TFLite	**TensorFlow Lite**

EPDS	**Edinburgh Postnatal Depression Scale**
WBS	**Work Breakdown Structure**
MVP	**Minimum Viable Product**

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1. INTRODUCTION

1.1 Background and Context

1.1.1 The Problem

Sri Lanka actually has one of the better maternal healthcare systems in South Asia. Compared to countries like India or Bangladesh, the infrastructure we have here through the Family Health Bureau, the network of government hospitals, and the community-level public health midwife system is genuinely impressive. Maternal mortality rates have come down a lot over the past few decades. But if you look carefully at what the system actually does, almost everything is focused on the physical side of pregnancy. Blood pressure, haemoglobin, baby's position, weight gain. That is what gets tracked at every antenatal clinic visit.

The mental and emotional side of pregnancy is basically ignored in the standard Sri Lankan antenatal care process. There is no structured screening tool for depression or anxiety built into the national protocol. Unless a woman herself brings up the fact that she is struggling emotionally, which most women here would not do because of social stigma around mental health, nobody asks. The doctor or midwife does their checks and moves to the next patient. The mental health burden a woman is carrying stays invisible.

And that burden is real. Pregnancy triggers a huge amount of emotional stress for many women. Worries about the baby's health, financial pressure, relationship difficulties, fear of childbirth, feeling alone when family is far away. These things do not go away just because nobody is asking about them. They accumulate between clinic visits, which in the Sri Lankan public system means accumulating for a whole month at a time. There is no touchpoint in the system for a woman to say she is not coping. The gap between appointments is where a lot of serious mental health problems start to take root.

1.1.2 Who Gets Affected

Every pregnant woman in Sri Lanka is potentially affected by this gap, but some groups are clearly more vulnerable than others. First-time mothers who have no idea what to expect emotionally from pregnancy are at high risk. Women going through a high-risk pregnancy where medical anxiety is constant are also particularly vulnerable. Women in plantation communities in the Central Province, women in the Northern and Eastern Provinces who are still recovering from the long-term trauma of the civil conflict period, and women in rural areas far from specialist mental health services all face this problem with fewer safety nets around them.

It is worth pointing out that this is not just about the mother's quality of life, although that should be reason enough. Research shows pretty clearly that sustained emotional stress during pregnancy has physical effects on the baby. Low birth weight, preterm delivery, and developmental problems in early childhood have all been linked to unmanaged maternal distress during pregnancy. So failing to address this is not just a gap in compassionate care. It has clinical downstream consequences that end up putting more pressure on the Sri Lankan health system later.

1.1.3 What the Numbers Say

A study carried out at the De Soysa Maternity Hospital in Colombo found that about 15.7 percent of pregnant women attending antenatal clinics showed clinically significant depressive symptoms based on Edinburgh Postnatal Depression Scale screening done at the second trimester [1]. Another study at Sri Jayewardenepura General Hospital looked specifically at anxiety in the third trimester and found prevalence rates of 22.8 percent [2]. A national level policy analysis by Senarath, Gunawardena and Jayawardena confirmed that psychological screening is completely absent from the standard antenatal care protocol across Sri Lankan public hospitals, at both primary and tertiary care levels [3].

On the technology side, the Telecommunications Regulatory Commission of Sri Lanka reported in their 2024 statistics that smartphone ownership among adults aged 18 to 45 is now at 67 percent nationally, with significant growth in rural provinces following the expansion of 4G coverage [4]. This means a mobile app can realistically reach pregnant women not just in Colombo but also in places like Anuradhapura, Badulla, and Trincomalee. The infrastructure is there to deliver something digital.

1.1.4 What Analysis of Existing Solutions and their limitations

Before deciding on this approach I looked at what is actually available and why none of it solves the problem for Sri Lankan pregnant women.

- The Sri Lankan public health system: The FHB provides printed antenatal health education booklets and midwife home visits. There is no digital emotional support tool and no mood screening during antenatal care. The system is built entirely around physical health monitoring and has no mechanism to support a woman emotionally between clinic visits.
- Global mental health chatbot apps like Woebot and Wysa: These are reasonably well designed for general anxiety and depression. But they are English only, have zero pregnancy-specific content, have no Sinhala support, and do not adjust their responses based on real-time emotional signals. A woman in Galle who feels more comfortable in Sinhala simply cannot use these apps effectively.
- General pregnancy apps like Glow and BabyCenter: Good for tracking physical milestones and fetal development information. They have essentially no emotional support features. They are information tools, not companions.
- Mood diary apps: Completely dependent on the user actively filling in how they feel. The women who most need support are the least likely to sit down and log their mood when they are in distress. These apps also have no adaptive response to what gets logged.
- Local telemedicine platforms like oDoc and ChannellingLK: These require booking a specific time, paying a consultation fee, and connecting with a doctor during working hours. They are not designed for ongoing emotional support and are not accessible in the middle of the night when a woman cannot sleep and is feeling overwhelmed.

So there is no tool in the current landscape that combines a Sinhala-capable interface, pregnancy-specific knowledge, passive mood detection, real-time adaptive responses, and offline functionality. That combination is what this component is built to provide.

1.1.5 The Proposed Solution

My part of this app is an AI chatbot built specifically for pregnant women in Sri Lanka. It talks to them in Sinhala and English, whichever they prefer or switch between, using language that feels familiar and warm rather than clinical. While the conversation is happening, a mood detection system runs in the background. Every 30 to 40 seconds, the front camera quietly takes a single photo of the user's face and passes it through a facial expression recognition model running on the phone itself. The result, whether she looks calm, a bit stressed, or clearly upset, feeds into a mood score along with signals from how she is typing, how long her responses are taking, and the emotional tone of her words.

When that mood score crosses a threshold that suggests she is significantly distressed, the chatbot changes how it talks to her. It becomes more gentle and reassuring, it recommends some calming Sinhala music or a breathing exercise, and it gently suggests she reach out to her doctor or midwife. All of this without her needing to say she is struggling or even realise the system has picked something up.

Everything is saved on her phone. Old conversations, saved music tracks, everything. It all works offline too, which matters a lot for someone in a rural area where the internet cuts out regularly.

1.1.6 SDG Alignment

- SDG 3 (Good Health and Well-Being): This component directly supports maternal mental health, which improves pregnancy outcomes for both mother and baby in the Sri Lankan context.
- SDG 5 (Gender Equality): Providing structured emotional support technology designed specifically around women's healthcare needs contributes to reducing the healthcare gap that Sri Lankan women currently face.
- SDG 10 (Reduced Inequalities): Women in rural areas and plantation communities who have limited access to mental health services can benefit from this just through their smartphone.

1.1.7 Research Cluster Alignment

Within SLIIT's research structure this project falls under the Intelligent Systems and Data Science cluster. It specifically touches on affective computing, multilingual NLP with a focus on Sinhala, and mobile AI applications for healthcare in the Sri Lankan setting. These are areas where there is very little existing work done with a Sri Lankan focus, so this contributes something genuinely novel to the local research landscape.

1.2 Literature Review

I read through a lot of material before settling on the technical approach for this component. I focused on research about Sri Lankan maternal mental health, facial emotion recognition, chatbot-based mental health support, Sinhala NLP, and multimodal emotion detection. Here is what I found and how it actually shaped the decisions I made.

Antenatal Depression and Anxiety in Sri Lanka. Rodrigo and Rajapakse ran a hospital-based study across two major government hospitals in the Western Province and found that 19.3 percent of pregnant women at antenatal clinics had significant depressive symptoms based on EPDS screening [1]. What really stood out from their paper was the point that almost all of these cases went completely unidentified during routine care because there is no screening protocol in place. It is not that the doctors did not care, there just was no system to catch it. A separate study by Fonseka, Wijesinghe, and Perera at Sri Jayewardenepura General Hospital looked at third-trimester anxiety specifically and found rates of 22.8 percent, with social isolation and financial stress being the strongest contributing factors among Sri Lankan women specifically [2]. Both of these papers argue directly for some kind of digital between-appointment intervention, which is exactly what I am building.

Mental Health Screening in Sri Lankan Antenatal Care. Senarath, Gunawardena, and Jayawardena published a policy analysis looking at the national Maternal and Child Health programme and confirmed what the above studies suggest, that psychological screening is entirely absent from the standard protocol at both primary and tertiary care levels in Sri Lanka [3]. They specifically recommended that mobile technology could be a practical delivery mechanism given rising smartphone penetration. That is a direct validation of this project's approach from a Sri Lankan policy research perspective.

Sri Lanka Smartphone and Internet Usage. The TRCSL annual statistical report for 2024 recorded active mobile subscriptions of 31.7 million against a national population of about 22 million, reflecting multiple SIM ownership, and smartphone usage among the 18 to 45 age group at 67 percent [4]. Growth has been particularly notable in provinces like Southern, North Western, and North Central, which are areas with large rural pregnant populations and limited specialist mental health access. The data makes it clear that reaching these women through a mobile app is genuinely feasible, not just in theory.

Facial Expression Recognition with Deep Learning. Mollahosseini, Hasani, and Mahoor published the AffectNet dataset paper in 2019, showing that CNN models trained on large diverse facial expression datasets reach accuracy above 65 percent for 8-class emotion classification under real-world conditions including varied lighting and partial occlusion [5]. For my simplified 3-class output the expected accuracy is higher than that. Their work was specifically done with real-world image conditions which is what I need since the camera will be capturing someone at home, not in a controlled lab setting.

Typing Behavior and Stress Detection. Giannakakis, Grigoriadis, and Tsiknakis showed in 2022 that typing behavioral signals from touchscreen input, things like inter-key timing and pause patterns, are statistically significantly different between stressed and non-stressed individuals [6]. This validates using typing speed and response delay as secondary mood indicators without needing any extra hardware or any action from the user. It just happens as part of normal chatting.

Chatbots for Perinatal Mental Health. Haga, Drozd, and Aas did a systematic review in 2021 looking at digital health interventions for perinatal depression and anxiety [7]. They found that while the number of apps in this space has grown, very few have been properly evaluated in clinical settings, and almost none of them use passive mood monitoring. They specifically called out the absence of apps targeting antenatal mental health in South and Southeast Asian contexts as a significant gap in the literature. This implementation for Sri Lanka is a direct response to that.

Music Therapy in Pregnancy, Sri Lankan Evidence. Sunitha and Nalini conducted a randomised controlled trial at a government maternity unit in Sri Lanka and found that listening to preferred music for 20 minutes per day reduced self-reported anxiety scores by a statistically significant amount over four weeks [8]. This is locally generated clinical evidence from Sri Lanka itself, not just a general finding from a Western study. It grounds the music recommendation feature in this component in a proper evidence base relevant to the actual user group.

Sinhala NLP. Wijeratne, Sheth, and Chandrasekar published work in 2019 on NLP tools for under-resourced languages including Sinhala, noting that while transformer models have been adapted for many South Asian languages, Sinhala remains significantly under-resourced in labeled datasets and pre-trained models [9]. The Language Technology Research Laboratory at the University of Moratuwa has produced some preliminary BERT-based work fine-tuned on Sinhala text that could support basic sentiment classification. Given this, my plan is to use a hybrid approach combining keyword-based scoring and model-based scoring for Sinhala text rather than relying solely on a model that may not yet be reliable enough.

Author / Year	Focus	Key Finding	What It Means for This Work
Rodrigo & Rajapakse (2019)	Antenatal depression, Sri Lanka	19.3% prevalence at Western Province hospitals, no screening in place	Confirms the core problem this component is solving
Fonseka et al. (2021)	Antenatal anxiety, Sri Lanka	22.8% at Sri Jayewardenepura GH,	Shapes the chatbot's understanding of local stressors

		social isolation main factor	
Senarath et al. (2020)	MCH policy analysis, Sri Lanka	Mental health absent from national antenatal protocol, mobile recommended	Direct policy validation for this project's approach
TRCSL (2024)	Smartphone stats, Sri Lanka	67% ownership, rural 4G expansion	Confirms app can reach women across Sri Lanka
Mollahosseini et al. (2019)	AffectNet FER dataset and CNN results	65%+ accuracy in real-world conditions	Basis for the on-device FER model used here
Giannakakis et al. (2022)	Typing behavior and stress	Passive touchscreen signals detect stress reliably	Validates typing speed and delay as secondary mood signals
Haga et al. (2021)	Perinatal mental health app review	No passive monitoring apps exist for South Asian context	This component fills exactly that gap for Sri Lanka
Sunitha & Nalini (2022)	Music therapy, Sri Lanka RCT	Music reduces antenatal anxiety, locally proven	Grounds music recommendation in Sri Lankan clinical evidence
Wijeratne et al. (2019)	Sinhala NLP resources	Sinhala is under-resourced, hybrid approach needed	Informs Sinhala sentiment analysis design decision

1.3 Research Gap

Gap 1: There is no Sinhala-language pregnancy emotional support chatbot built for Sri Lankan women.

Every mental health chatbot that exists right now is in English and designed around Western healthcare norms. Nothing has been built specifically for a Sri Lankan pregnant woman chatting in Sinhala with culturally familiar language. This is not just a nice-to-have. A woman who is not

comfortable in English will not use an English-only app when she is emotionally vulnerable. Building this in Sinhala is what makes it actually usable for the majority of Sri Lankan pregnant women.

Gap 2: Passive multimodal mood detection has not been applied in any pregnancy support app for Sri Lankan or South Asian women.

The literature shows each piece of this puzzle separately. Facial expression recognition has been studied. Chat behavior as a stress signal has been studied. Text sentiment analysis has been studied. But combining all three into a unified passive mood pipeline specifically designed to run during a chatbot conversation, without requiring any input from the user, has not been done in any pregnancy support application in this region.

Gap 3: Locally validated therapeutic music recommendation based on real-time mood detection does not exist in the Sri Lankan maternal health app space.

Sunitha and Nalini have shown in a Sri Lankan clinical trial that music reduces antenatal anxiety. But no app currently uses that evidence to build a real-time mood-responsive music recommendation feature for Sri Lankan pregnant women. That combination of local clinical evidence plus working technology implementation is what this component contributes.

2.OBJECTIVE

2.1 Main Objective

To design and build an AI-powered chatbot module within the MaternaLink pregnancy support application that tracks the emotional wellbeing of pregnant women in Sri Lanka in real time through passive facial expression detection and chat behavior monitoring, responds in Sinhala and English with conversation that adjusts to how she is feeling, and provides culturally appropriate therapeutic music recommendations grounded in Sri Lankan clinical evidence, with all chat history and content stored locally for full offline access.

2.2 Specific Objectives

Building a Bilingual Chatbot Interface That Feels Right for Sri Lankan Women. The first thing I need to actually produce is a chatbot screen that a pregnant woman in Sri Lanka would genuinely want to use. It has to work in Sinhala Unicode and English, the interface has to be warm and soft rather than clinical, and the language the chatbot uses has to feel natural and respectful in the local cultural context. Not formal, not robotic. All conversations get saved on the device locally.

Implementing On-Device Facial Emotion Recognition. Every 30 to 40 seconds while the chat is active, the front camera takes a single frame and it gets processed through a TensorFlow Lite facial expression model running entirely on the device. No image ever leaves the phone. The model classifies the expression into one of three categories: calm, mildly distressed, or significantly distressed. The TFLite format is important because it runs on mid-range Android phones, which is what most people in Sri Lanka are using.

Building the Chat Behavior and Sentiment Analysis Module. Alongside the facial analysis, the system passively tracks typing speed, response time, and the emotional tone of what the user is typing. For Sinhala text I am using a hybrid approach since proper Sinhala NLP models are still limited, combining a keyword-based scoring method with whatever model-based support I can get from the University of Moratuwa's Sinhala BERT work. For English I am using a pre-trained HuggingFace sentiment model via an API call.

Creating the Mood Fusion Engine and Adaptive Response System. The three signals from the face, the text, and the behavior all get combined in a weighted formula to produce a single mood score. That score gets passed to the chatbot as context so the language model knows how to adjust its tone. When the score goes above the distressed threshold, the system prompt tells the model to respond more gently and the music recommendation appears.

Integrating Culturally Appropriate Therapeutic Content and Local Offline Storage. The music library has to include content that actually resonates with Sri Lankan women, instrumental versions of familiar Sinhala melodies and devotional music that is locally associated with calm and comfort. Everything the user saves gets stored locally. The chat history is also stored locally and accessible without internet, which is crucial for women in parts of the country where connectivity is not reliable.

3.METHODOLOGY

3.1 Research Approach

I am using Design Science Research for this because the whole point is to produce a working system and then evaluate whether it actually solves the real problem for the target users. DSR fits well because it is not just about building something and saying it is done. It is about building, evaluating against real criteria, and refining. Development follows Agile with two-week sprints so I can test things in parts rather than discovering problems only at the very end. Each sprint is supposed to produce something I can actually run and look at.

3.2 Technical Approach

3.2.1 The Bilingual Chatbot Interface

The chatbot screen is part of the React Native app that the whole MaternaLink group is building together. I am responsible for this particular screen and its supporting backend. The UI is designed to look and feel like a messaging app, something Sri Lankan women are already familiar with from using WhatsApp and Viber every day. The input field supports Sinhala Unicode keyboard input on Android without any extra configuration needed from the user.

The language the chatbot uses is something I am spending real time on. It cannot just be a translation of English phrases into Sinhala. It has to use the right register, the right level of formality, the right way of addressing someone that feels warm and appropriate in a Sri Lankan context. I am planning to get a Sinhala language consultant, ideally someone with a health communication background, to review the chatbot script and the model's system prompt before testing. The English version also needs to be simple and warm, not academic.

3.2.2 Facial Emotion Recognition Running On-Device

The camera capture happens as a background task using the Expo Camera API in React Native. A timer fires every 30 to 40 seconds while the chat screen is active. One frame is captured, resized to 48 by 48 pixels, converted to grayscale, and normalised. This all happens on the device before anything gets passed to the model. The frame then goes into the TensorFlow Lite FER model which produces a probability for each of the three mood categories.

The model itself is a MobileNetV2 architecture pre-trained on ImageNet and fine-tuned on FER-2013 and AffectNet data, then converted to TFLite format and set to a 3-class output. I chose MobileNetV2 because it is lightweight enough to run on a budget Android phone without killing the battery or slowing down the chat. The FER-2013 dataset has about 35,000 labeled facial expression images across seven categories. I am mapping those seven down to three for this application. Fine-tuning rather than training from scratch means I need a lot less data and the model converges much faster.

The whole point of running inference on the device is that no facial image ever gets stored anywhere or sent to a server. This addresses a real concern for Sri Lankan users who are

generally cautious about privacy and data sharing, particularly for something as personal as photos taken of their own face.

3.2.3 Chat Behavior and English Sentiment Analysis

Three passive signals come from the chat behavior itself. Typing speed is measured in characters per second as the user is composing each message. Response delay is the time in seconds between when the chatbot sent its last message and when the user starts typing. A consistently long response delay, or typing that is very slow and hesitant, is treated as a possible signal of low mood or disengagement. These thresholds are simple but they add useful context when combined with the other signals.

For text sentiment, I am handling English differently. For English messages a lightweight pre-trained sentiment classifier from HuggingFace gets called via a backend API. It is fast and reliable for English. For Sinhala messages I am taking a hybrid approach. I am building a keyword list of emotionally significant Sinhala words relevant to pregnancy and mental state, words relating to fear, sadness, worry, happiness, exhaustion, and hope, and using that for a base score. On top of that I am integrating the preliminary Sinhala BERT model from the University of Moratuwa's Language Technology Research Lab as a supplementary classifier. The two scores get combined with the keyword result carrying more weight for now given the current reliability limitations of the Sinhala model. This will be adjusted during testing based on actual accuracy.

3.2.4 Mood Fusion Engine

The fusion layer takes the most recent output from each of the three streams. The facial expression result carries 50 percent of the weight because it is the most direct emotional signal. Text sentiment carries 35 percent. Behavioral signals carry the remaining 15 percent. These get combined into a single mood score from 0 to 100 that maps to one of the three categories.

Every time the user sends a message, this score is recalculated and the chatbot context is updated. The backend Node.js service that handles the LLM API call receives the current mood category along with the message and injects it into the system prompt so the language model knows how to pitch its response. When the mood is significantly distressed the system prompt specifically tells the model to use softer language, avoid giving information-heavy responses, gently acknowledge that she seems to be going through something hard, and suggest she reach out to her midwife or doctor. The music recommendation banner also triggers at this point.

3.2.5 Therapeutic Content and Offline Storage

The music library is curated specifically for Sri Lankan pregnant women. It includes instrumental versions of familiar Sinhala devotional melodies, soft instrumental folk music, and general calm instrumental music. All tracks are royalty-free or locally licensed. The audio player is built into the app using a React Native audio library. When the user saves a track it goes into AsyncStorage on the device. She can play it any time without internet.

Chat history is also written to AsyncStorage after every message exchange. The history screen shows past conversations sorted by date, with a preview of the first message and the final mood indicator for that session. The Sinhala text renders correctly in the history view. Everything loads from local storage only with no network call needed. This offline capability is not an optional extra, it is a core requirement for reaching women in Uva Province, Vanni, or the plantation districts of the Central Province where connectivity genuinely cannot be relied on.

3.3 Tools and Platforms

Tool	Purpose	Why This Choice for Sri Lanka
React Native (Expo)	Mobile chatbot UI and camera access	Runs on Android and iOS; Expo Camera API handles front camera; works on mid-range Android phones common in Sri Lanka
TensorFlow Lite	On-device FER model inference	Runs offline on budget Android devices; no server needed; keeps all facial data on-device for privacy
FER-2013 + AffectNet	Training data for FER model	Two largest open-access labeled facial expression datasets; diverse enough for real-world home conditions
Sinhala BERT (UoM LT Lab)	Sinhala sentiment analysis	Best currently available Sinhala NLP model; produced by University of Moratuwa
HuggingFace Transformers	English text sentiment analysis	Large library of fast pre-trained sentiment models suitable for API inference
Python + Keras	FER model training and TFLite export	Standard deep learning stack; native MobileNetV2 support in Keras

AsyncStorage (React Native)	Local chat and music offline storage	Built into React Native; fully offline; no server dependency for core features
Node.js + Express	Backend API for LLM with mood context	Lightweight; easy to inject mood context into LLM system prompt
Google Colab Pro	FER model training environment	GPU access at low cost; free tier too slow for full MobileNetV2 training

3.4 Data and Ethics

There are two categories of data in this project that need careful handling and both of them are relevant to the Sri Lankan privacy context.

For training data, I am using FER-2013 and AffectNet which are both publicly available for research use under open licenses. I am not collecting facial images from any real user for training. No Sri Lankan woman's face is being used to train this model.

For live user data, the only facial data that gets processed is the single camera frame taken every 30 to 40 seconds during a chat session. That frame is preprocessed and run through the TFLite model entirely on the device. The raw image is discarded immediately after inference. It is never stored on the phone and never sent to any server. This is an important privacy guarantee given how sensitive it is to capture photos of someone's face.

Chat messages sent to the LLM backend are anonymized before leaving the phone. No name, no phone number, no user ID is attached to any API request. Sinhala text processed for sentiment analysis goes through the same anonymization step. Chat history stored locally on the device is in encrypted AsyncStorage.

The privacy notice will appear in both Sinhala and English the first time a user opens the chatbot, before any conversation begins. It will explain in plain language what data is collected, that photos are never stored or transmitted, and that all facial analysis happens on the phone. There will be a clear one-tap toggle to disable camera access at any time, in which case the system continues with just text and behavioral signals.

I will submit a formal ethics application to the SLIIT Institutional Ethical Review Committee before any user testing begins. All test participants will be adults who have given written informed consent. Given the sensitive combination of pregnancy and mental health in the Sri Lankan cultural context, I will be particularly careful to ensure no participant feels uncomfortable or pressured during testing sessions.

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3.5 How I Will Know If It Works

What I Am Measuring	Target	How I Will Measure It
FER model accuracy on 3-class output	At least 65%	Precision, recall, and F1 score on a held-out test split from FER-2013 and AffectNet
On-device FER inference time	Under 3 seconds per frame	Timing logs recorded on a mid-range Android device during test sessions
Mood detection agreement with self-report	70% or above agreement	Ask test participants to rate their own mood after session; compare with system output
User satisfaction with chatbot tone (Sinhala)	4 out of 5 or above average	Post-session questionnaire in Sinhala with 10 Sri Lankan participants
Sinhala sentiment analysis accuracy	At least 70% on test phrases	Manually label 200 Sinhala pregnancy-related phrases and measure classifier accuracy
Offline functionality: chat history and music	100% accessible without internet	Test by putting device in airplane mode and verifying full access to stored data

4.HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

The chatbot module is a semi-independent component inside the MaternaLink app. It has four layers that work together: the bilingual chatbot user interface that the woman interacts with, the background mood detection pipeline that runs passively while she chats, the mood fusion and adaptive response engine that connects the detected mood to how the chatbot talks, and the local offline storage layer that keeps everything on her device. The first three layers work together during an active conversation. The storage layer is always running in the background saving everything as it happens.

4.2 Layer-by-Layer Description

Layer 1 The Chat Interface

The screen looks like a standard messaging app. Chat bubbles on the left for the chatbot, on the right for the user. The input bar at the bottom accepts Sinhala Unicode and English text. There is a small soft-colored indicator at the top showing the current mood state, green for calm, yellow for mildly distressed, red for significantly distressed, but it is subtle enough that it does not feel alarming or clinical. When the music recommendation triggers, a soft banner slides up above the input bar with a track name and a play button. The whole visual design is warm and calm, nothing that looks like a hospital app.

Layer 2 Background Mood Detection Pipeline

This runs entirely without the user needing to do anything. A timer fires every 30 to 40 seconds. It triggers the front camera capture, a single frame, processed on-device through the TFLite FER model. In parallel, every message the user sends goes through the sentiment classifier, and the timestamps of all messages are used to calculate typing speed and response delay. The results of all three streams are held in memory and passed to the fusion layer every time the user sends a message.

Layer 3 Mood Fusion and Adaptive Response Engine

The fusion layer takes the three latest signals and computes the weighted mood score. That score updates the chatbot context. The Node.js backend receives the user's message along with the current mood category and calls the LLM API with a system prompt that includes the mood context. The LLM returns a response that has been shaped by knowing whether the user is calm or distressed. If the mood is significantly distressed, the system prompt change is more substantial and the music recommendation banner is triggered on the front end.

Layer 4 Local Offline Storage

Every message exchange is written to AsyncStorage immediately. Music the user saves goes to the same local storage. The history screen and the music screen both read entirely from local storage with no network call. For a woman in a rural area of Sri Lanka where mobile data is unreliable, this means the app is fully functional even when she is offline. She can read old conversations, listen to saved music, and the chatbot will catch up and sync any new messages the next time she connects.

4.3 Data Flow — Step by Step

1. User opens the chatbot screen and types a message in Sinhala or English.
2. Message is saved to local AsyncStorage immediately and sent to the Node.js backend with the current mood context.
3. Backend formats the full request including the system prompt updated with the mood context and calls the LLM API.
4. LLM returns a response with tone adjusted to match the current mood category.

5. Response is displayed in the chat and saved to local storage.
6. Background timer fires. Front camera captures a single frame silently.
7. Frame is preprocessed on-device and passed through TFLite FER model. Result updates the facial mood signal in memory.
8. Fusion layer recalculates the combined mood score using the latest FER result, text sentiment, and behavioral signals.
9. Updated mood score is available for the next message exchange.
10. If mood score crosses the significantly distressed threshold, music banner triggers and the next chatbot response uses the gentler distressed-mode system prompt.

5.USER REQUIREMENTS

5.1 Stakeholders

- Pregnant Women in Sri Lanka: The main users. They range from urban Colombo to rural Uva Province. They chat in Sinhala or English, get mood-aware responses, receive music suggestions, and access their history offline.
- Doctors and Public Health Midwives: They can see a mood history summary through the doctor-facing side of MaternaLink. This gives them something they have never had before, a picture of how their patient has been feeling between clinic visits.
- System Administrators: Monitor model accuracy over time, update chatbot language content, manage the Sinhala music library.

5.2 Functional Requirements

1. FR1: The chatbot must support conversation in Sinhala Unicode and English with the user able to switch between languages at any point in the conversation.
2. FR2: The system must silently capture a front camera frame every 30 to 40 seconds during an active chat session and classify it using the on-device TFLite FER model without interrupting the conversation.
3. FR3: The system must passively track typing speed, response delay, and text sentiment for every message and combine these with the FER result to produce a current mood classification.
4. FR4: The chatbot response tone must adjust based on the current mood classification within the same chat session without any action required from the user.
5. FR5: The system must display a music recommendation and trigger the gentler response mode when the mood classification reaches significantly distressed.

6. FR6: All chat history in both Sinhala and English must be saved locally on the device and must be fully accessible without internet.
7. FR7: Music saved by the user must be stored locally and playable in offline mode.
8. FR8: The user must be able to disable camera access via a single toggle, and mood detection must continue using only text and behavioral signals in that case.
9. FR9: A mood history summary must be made available to the doctor-facing component of MaternaLink.
10. FR10: The privacy notice must display in both Sinhala and English before first use.

5.3 Non-Functional Requirements

- Performance: FER inference must complete within 3 seconds per frame. Chatbot response must appear within 5 seconds on a typical Sri Lankan 4G connection.
- Privacy: No facial image is ever stored or transmitted. All FER processing is on-device. This must be clearly stated in Sinhala and English.
- Language: Sinhala Unicode must render correctly in all chat bubbles, history, and music labels on Android.
- Compatibility: Must run on mid-range Android devices which are the dominant smartphone type in Sri Lanka.
- Offline: Core functions including history and music playback must work completely without internet.
- Usability: Language must feel natural and respectful in the Sri Lankan cultural context. Tested specifically with Sri Lankan participants.

6.COMMERCIALIZATION PLAN

6.1 Target Market

The primary market is pregnant women across Sri Lanka. The Department of Census and Statistics records approximately 340,000 registered births per year in Sri Lanka, meaning there are roughly that many pregnant women in the country at any point in time [12]. With 67 percent smartphone ownership in the relevant age group and growing, the addressable market through a mobile app is substantial even before factoring in any institutional partnerships. Initial launch would focus on the Western, Southern, and Sabaragamuwa Provinces where smartphone use is highest, with a planned rollout to the Northern, Eastern, Uva, and North Central Provinces.

Secondary markets include Sri Lankan government hospitals and private maternity hospitals that want to offer digital antenatal support to their patients, maternal health NGOs operating in Sri Lanka like the Family Planning Association of Sri Lanka and Sarvodaya's health program, and potentially the Ministry of Health and the Family Health Bureau as an institutional partner for integration into the national antenatal care system.

6.2 Value Proposition

- The only Sinhala-language pregnancy emotional support chatbot designed specifically for Sri Lankan women's cultural context and healthcare setting.
- Passively monitors emotional state through facial expression and chat behavior so it catches distress even when the woman does not say anything.
- Works fully offline for women in rural and plantation areas where connectivity is not reliable.
- Gives doctors and midwives a picture of their patient's emotional health between appointments, something completely missing from current Sri Lankan antenatal care.
- Music recommendation feature grounded in a Sri Lankan clinical trial, not just international evidence.

6.3 Revenue Model

- Free app on Google Play Store: Basic chatbot and mood tracking free for all users to drive adoption in the Sri Lankan market.
- Premium subscription at LKR 500 per month: Extended chat history, detailed mood trend reports, and larger Sinhala and English music library.
- Institutional licensing for hospitals and clinics: Monthly fee starting at LKR 15,000 for facilities up to 200 active patients, scaled for larger institutions.
- Government partnership: Ministry of Health or FHB license to offer the chatbot to all registered antenatal patients nationally at a negotiated public health rate.
- Anonymized aggregate data insights: District-level maternal mental health trend data licensed to academic researchers and public health institutions under strict ethical guidelines.

6.4 Cost Estimation

Item	Estimated Cost (USD / year)
Backend hosting for Node.js API and LLM calls	**$ 3,600**
LLM API usage fees	**$ 2,400**
FER and Sinhala NLP model training (cloud GPU)	**$ 800**
Royalty-free Sinhala and instrumental music licensing	**$ 600**
Sinhala language and cultural content review	**$ 1,000**
Part-time maintenance developer	**$ 5,000**
Marketing and community outreach in Sri Lanka	**$ 1,500**
Total	**$ 14,900 / year**

6.5 Pricing Strategy

Launch free to build a user base and generate real usage data to show the app works before asking any institution to pay. The LKR 500 per month premium tier is priced to be affordable for middle-income Sri Lankan women while covering ongoing API costs. Hospital and clinic pricing is set at a level realistic for a Sri Lankan healthcare facility budget. The government partnership route is the most impactful long-term pathway and is worth investing in a fully free pilot programme to prove outcomes before any commercial negotiation.

6.6 Competitive Advantage

- Only Sinhala-language pregnancy emotional support tool built for the Sri Lankan market.
- Passive multimodal mood detection combining facial analysis and chat behavior, not available in any existing product.
- Full offline capability for core features, essential for the rural Sri Lankan context.
- Mood data feeds into the doctor-facing component of MaternaLink creating an end-to-end care loop.

6.7 Intellectual Property

All code, trained model weights, and the mood fusion algorithm will be copyrighted under SLIIT. The combination of passive bilingual facial monitoring with Sinhala-English chat behavior analysis for pregnancy-specific mood detection may qualify for a provisional patent. A dual-license model is

planned: MIT open-source for academic and non-profit use, commercial license for revenue-generating deployments by hospitals and institutions.

7.BUDGET AND JUSTIFICATION

#	Item	Cost (LKR)	Justification
1	Google Colab Pro – 3 months for FER model training	7,500	Free GPU tier is too slow for full MobileNetV2 fine-tuning runs on FER-2013
2	LLM API access for chatbot backend – development and testing quota	10,000	Required to run and test the chatbot during development; cannot avoid this cost
3	Royalty-free Sinhala and instrumental music library	5,000	Therapeutic music must be properly licensed; sourcing appropriate Sinhala content locally requires a budget
4	Sinhala language and cultural review by a native speaker consultant	4,000	Essential to ensure chatbot language feels natural and appropriate for Sri Lankan women; cannot do this without a native speaker review
5	Domain name and SSL certificate for backend API – 1 year	3,500	Required for HTTPS deployment of the Node.js backend
6	AWS backend hosting beyond free tier – 3 months	8,000	Free tier covers early development but not realistic load testing or the demo phase
7	React Native, TensorFlow Lite, Keras, HuggingFace libraries	0	All open-source; no cost

8	FER-2013 and AffectNet datasets	0	Both publicly available under open research licenses
9	Report printing and binding for final submission	2,000	Physical submission requirement for the final dissertation
10	Contingency buffer – 10% of total	4,000	Covers unexpected API cost spikes or tooling changes during development
	TOTAL	44,000 LKR	Approximately USD 135 at current exchange rate

8.WORK BREAKDOWN STRUCTURE (WBS)

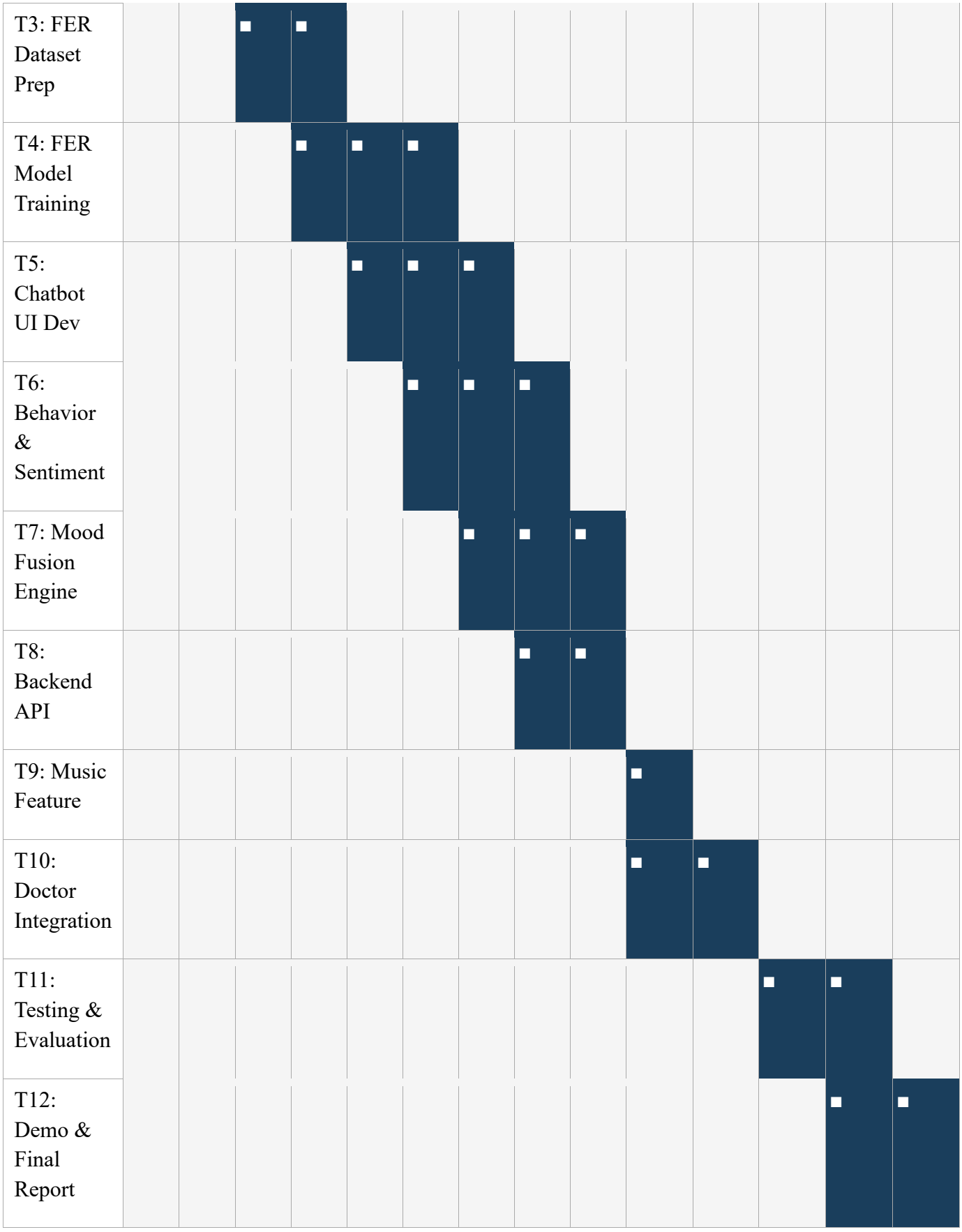
7.1 Task Breakdown — Livera S.D (IT22638168)

ID	Task	Deliverable	Timeline
T1	Literature review on Sri Lankan maternal mental health studies and research gap analysis	Sections 1.2 and 1.3 of proposal with Sri Lankan references	Weeks 1–2
T2	Chatbot UI wireframe with Sinhala/English bilingual layout design and local storage schema	Figma wireframes and storage structure diagram	Week 3
T3	FER model dataset prep: FER-2013 and AffectNet preprocessing, 3-class remapping pipeline	Preprocessed dataset and augmentation pipeline ready	Weeks 3–4
T4	FER model training with MobileNetV2 transfer learning and TFLite export for Android	Trained .tflite model with evaluation accuracy report	Weeks 4–6
T5	React Native chatbot UI build with Sinhala Unicode rendering and front camera capture integration	Working bilingual chatbot screen with background camera capture	Weeks 5–7
T6	Chat behavior analysis: typing speed, response delay tracking, and bilingual sentiment pipeline	Working module producing per-message mood signal inputs	Weeks 6–8
T7	Mood fusion engine: weighted signal combination, mood scoring, adaptive tone context injection	Fusion layer outputting mood score, chatbot tone adjusting correctly	Weeks 7–9

T8	Node.js backend API with LLM integration: bilingual mood context injection and Sinhala response support	Functional chatbot backend responding in Sinhala and English per mood	Weeks 8–9
T9	Sinhala therapeutic music library curation, local licensing, audio player and offline playlist feature	Working player with locally stored Sinhala and instrumental tracks	Week 10
T10	Integration with doctor-facing mood summary component of MaternaLink	Mood history readable from the doctor side of the app	Weeks 10–11
T11	Full end-to-end testing: FER accuracy, Sinhala rendering, offline mode, Sri Lankan user testing	Test report with all metrics from Section 3.5 completed	Weeks 12–13
T12	Demo preparation and final dissertation chapter writing	Working live demo on Android and completed written submission	Weeks 13–14

9.GANTT CHART

	Semester 1						Semester 2							
Task	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**	**W9**	**W10**	**W11**	**W12**	**W13**	**W14**
T1: Literature Review	■	■												
T2: UI & Storage Design			■											


T3: FER Dataset Prep
T4: FER Model Training
T5: Chatbot UI Dev
T6: Behavior & Sentiment
T7: Mood Fusion Engine
T8: Backend API
T9: Music Feature
T10: Doctor Integration
T11: Testing & Evaluation
T12: Demo & Final Report

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11.APPENDICES

11.1 Appendix A Survey Instruments

Before I started building anything I wanted to actually understand how pregnant women in Sri Lanka deal with emotional stress and whether they would genuinely use a Sinhala chatbot app. I designed two instruments: a pre-development survey targeting Sri Lankan women specifically, and a usability testing script for when the app is built.

A.1 Pre-Development User Survey

Who: Pregnant women or women who have been pregnant within the last two years in Sri Lanka. Targeting participants from at least five districts including both urban (Colombo, Kandy) and rural (Matara, Anuradhapura, Badulla) areas.

How: Google Form in Sinhala and English, shared through Sri Lankan maternal health Facebook groups, hospital WhatsApp groups for antenatal patients, and FHB community health worker networks.

Target: 40 to 60 responses.

1. Which district in Sri Lanka do you currently live in?
2. Are you currently pregnant or have you been pregnant in the last two years? (Yes / No)
3. During your pregnancy, did you ever feel anxious, sad, stressed, or emotionally overwhelmed? (Yes / Sometimes / No)
4. When you felt that way, who or what did you turn to? (tick all that apply) – Husband or family / Doctor or midwife / Prayer or religious practice / Tried to manage alone / Used an app or searched online / Other
5. Do you own a smartphone? (Yes / No)
6. Which language do you prefer for everyday communication? (Sinhala / English / Both equally)
7. Would you feel comfortable talking to an AI chatbot about how you are feeling during pregnancy? (Yes / Maybe / No – Why not?)
8. If the app accessed your front camera briefly every minute while you chat to better understand your mood, but all the analysis happened only on your phone and no photos were ever stored or sent anywhere, would you be okay with that? (Yes, if clearly explained / Maybe / No)
9. What kind of support would you most want from a pregnancy app? (tick all that apply) – Someone to talk to in Sinhala / Calming music suggestions / Pregnancy information / Clinic visit reminders / Something else
10. Is there anything that would stop you from using something like this? (open text in Sinhala or English)

A.2 Usability Testing Script

Who: 10 volunteers, at least 5 of whom are female and ideally some with personal experience of pregnancy or close family experience.

How: One-on-one moderated session of 25 to 30 minutes. Interface shown in Sinhala by default. I will not give hints unless the participant is completely stuck.

- Task 1: Open the chatbot and start a conversation in Sinhala about how you are feeling today.
- Task 2: Keep chatting for at least 5 minutes while I observe.
- Task 3: Switch to English at some point and continue the conversation.
- Task 4: Find the music recommendation that appeared and try playing it.
- Task 5: Save the track to your local playlist.
- Task 6: Turn off your internet connection and verify that your chat history and music are still accessible.

After the tasks I will ask (in Sinhala or English depending on the participant):

11. Did the chatbot feel like it understood how you were feeling? (1 = Not at all, 5 = Completely)
12. Did the language feel natural and respectful in a Sri Lankan context?
13. Did you notice the chatbot's way of talking change at any point?
14. Would you recommend this to a pregnant friend or family member?
15. What one thing would you change?

11.2 Appendix B Detailed Diagrams

B.1 **Local Storage Schema (AsyncStorage on Device)**

Field	Type	What It Stores
session_id	**UUID**	**Unique identifier for each chat session**

timestamp	**ISO String**	**Date and time when the session started**
language	**String**	**Primary language of the session: sinhala or english**
messages	**JSON Array**	**All messages with sender label, text content, and timestamp**
mood_history	**JSON Array**	**Mood classifications recorded during the session with timestamps**
final_mood	**String**	**Overall mood at end of session: calm / mildly_distressed / significantly_distressed**
music_playlist	**JSON Array**	**Saved music tracks with title, file path, language tag, and date saved**
camera_enabled	**Boolean**	**Whether the user allowed camera access for mood detection this session**

B.2 **FER Model Architecture Summary**

Layer	What It Does	Output Shape
Input	Receives the 48x48 greyscale frame captured from front camera	48 × 48 × 1
MobileNetV2 base (frozen first, partially unfrozen later)	Extracts facial feature maps using ImageNet pretrained weights	3 × 3 × 1280
Global Average Pooling	Reduces spatial feature maps to a single vector	1,280
Dense layer (256 neurons, ReLU)	Learns classification-relevant combinations of features	256

Dropout (rate 0.3)	Reduces overfitting on the FER training data	256
Output Dense (3 neurons, Softmax)	Produces probability for calm / mildly distressed / significantly distressed	3

11.3 Appendix C — Preliminary UI Mockups

Proper wireframes will be done in Figma in Week 3. Here is the intended layout for the two main screens so the structure is clear before development starts.

Screen 1: Chatbot Main Screen (Sinhala Default). The background is a warm off-white. Chat bubbles for the chatbot are on the left in a soft teal color. The user's messages are on the right in white with a subtle shadow. A language toggle sits at the top right so the user can switch between Sinhala and English at any point. The mood indicator dot is in the top left corner, small and subtle, green, yellow, or red. The bottom input bar supports Sinhala Unicode keyboard input on Android. When a music recommendation triggers, a soft banner slides up above the input bar with the track name in Sinhala and a small play button. The whole design is warm and calm, nothing clinical or alarming.

Screen 2: History and Music Screen. Two tabs at the top: Chat History and My Music. The chat history tab shows conversations sorted newest first. Each entry shows the date, the language used, a preview line of the first message, and a small mood indicator dot. Tapping opens the full conversation. The music tab shows the locally saved playlist. Play, pause, and delete buttons are on each track. Everything loads from local storage only. No internet needed. Sinhala text renders correctly in both tabs.

11.4 Appendix D — Risk Analysis

I tried to think realistically about what could go wrong rather than being too optimistic. Here are the main risks and what I plan to do about each one.

#	Risk	Likelihood	Impact	What I Will Do
R1	FER model accuracy stays below 65% on 3-class output	Medium	High	Try heavier data augmentation first, then switch to EfficientNet-Lite as backup; test under varied Sri Lankan home lighting conditions specifically
R2	User denies camera access, breaking half the mood pipeline	Medium	Medium	Design graceful degradation from the beginning: system continues with text and behavior signals only; camera-off mode is a first-class path not an afterthought
R3	Sinhala sentiment analysis too inaccurate to be useful	Medium	High	Start with keyword-based scoring only; integrate UoM BERT model progressively as I validate its accuracy on pregnancy-specific Sinhala text
R4	LLM API costs exceed budget during development	Medium	Medium	Set hard spending limits on the API account from day one; have a local open-source LLM fallback ready if costs spike
R5	Mood fusion produces inconsistent results and chatbot tone shifts feel jarring	Medium	High	Use conservative thresholds: only shift tone when mood signal is clearly and consistently distressed across multiple consecutive readings
R6	Users uncomfortable with camera access	High	High	Very clear Sinhala and English consent screen before first use; visible indicator when camera is

	given Sri Lankan privacy culture			active; single-tap disable toggle always available
R7	Sinhala Unicode not rendering correctly on some Android devices	Medium	Medium	Test on at least five different Android device models available in the Sri Lankan market from the beginning of development
R8	Doctor integration component not ready when I need to connect	Low	Medium	Build mood summary export as a standalone JSON file first so it works independently of the other component being ready
R9	FER model biased toward lighter skin tones in training data	Medium	High	Deliberately test with participants of diverse Sri Lankan ethnicities during usability testing; apply skin-tone-aware augmentation during fine-tuning

11.5 Appendix E — Ethical Documents

E.1 How I am handling data in this project

There are two types of data in this project that need careful handling.

Training data: FER-2013 and AffectNet are both publicly available under open research licenses. I am not collecting facial images from any Sri Lankan woman for training purposes. No real user's face is involved in building the model.

Live user data: The only facial data that gets processed is the single camera frame taken every 30 to 40 seconds during a chat session. This is processed entirely on the phone using TFLite. The raw image is discarded immediately after the model produces its output. It is never stored anywhere on the device and never transmitted. The only output that persists is the mood category label, not the image.

Chat messages sent to the LLM backend are anonymized before leaving the phone. No name, no phone number, no device identifier is attached to any API request. Chat history stored locally is in encrypted AsyncStorage. The doctor-facing mood summary contains only aggregate mood categories and timestamps, no raw conversation text.

E.2 Privacy notice that will appear in the app

This is the draft privacy notice users will see in Sinhala and English before the chatbot opens for the first time:

"This chatbot is here to support you during your pregnancy. While you are chatting, this app may briefly use your front camera about once per minute to better understand how you are feeling. This analysis happens only on your phone. No photos are ever saved or sent anywhere. Your chat messages are used to generate responses but are not linked to your identity. All your conversations and saved music are stored only on this device. You can turn off camera access at any time in the settings. By continuing, you agree to these terms. Please read the full privacy policy at [link]."

The same notice will be displayed in Sinhala for users whose device language is set to Sinhala

E.3 Ethics clearance process

Before I run any user testing, including the usability sessions described in Appendix A.2, I will submit a formal ethics application to the SLIIT Institutional Ethical Review Committee. The application will cover exactly what data is being collected, how it is stored, who can access it, and how participants are giving informed consent.

All usability test participants will be adults who have volunteered to take part. Before each session starts, each person will read and sign a written consent form in Sinhala or English, whichever they prefer, confirming that participation is voluntary and that they can stop at any time. I will not collect any data from anyone under 18. All consent forms and session notes will be stored in a password-protected folder and deleted after the project is submitted and assessed.

Research Team:

Role	Student ID	Contribution
Health Monitoring and AI Risk Classification	IT22557506	Heart rate, BP, and blood sugar analysis; AI-driven risk level classification; automated alerts to doctor when risk is high

Role	Student ID	Contribution
CTG Scan AI Analysis and Report Generation	IT22272386	AI-based analysis of CTG scan images and values; automated clinical findings with doctor review and annotation capability
AI Chatbot with Mood Detection (This Component)	IT22638168	Bilingual Sinhala/English chatbot; facial emotion recognition; chat behavior analysis; mood fusion; adaptive responses; therapeutic music; offline storage
AI Recipe Suggestion from Daily Ingredients	IT22284716	Daily ingredient input from user; AI-generated Sri Lankan-appropriate meal suggestions; calorie and nutritional benefit analysis

Facilities and Resources:

Facility	Use
SLIIT IT Laboratory, Malabe	Main development environment for React Native app, Node.js backend, Android device testing, and Sinhala Unicode rendering verification
Google Colab Pro	FER model training with MobileNetV2 fine-tuning on FER-2013 and AffectNet; TFLite conversion and export
Personal Android Test Devices	Testing facial expression capture and TFLite inference under real Sri Lankan home lighting conditions on mid-range devices
University of Moratuwa Language Technology Lab (Online Resources)	Accessing Sinhala BERT model weights and documentation for the Sinhala sentiment analysis component
Team Members' Laptops	Day-to-day development, Python model work, and documentation outside lab hours

— End of Proposal Document —

IT22638168 —Pregnancy Support System

Department of Information Technology, SLIIT — January 2026